

## Coreset Subsampling

Given  $N$  training samples with soft labels  $Y \in \mathbb{R}^{N \times K}$  and domain assignments  $D \in \mathbb{Z}^N$ , we select a coreset of size  $\lfloor \alpha N \rfloor$  (default  $\alpha = 0.25$ ) using one of two strategies. Subsampling is performed independently per domain, with quota proportional to domain size.

### Strategy 1: Entropy-Stratified Sampling

1. Compute prediction entropy per sample:

$$H_i = - \sum_{k=1}^K y_{ik} \log y_{ik} \quad (1)$$

2. Assign each sample to a stratum  $(c_i, b_i)$  where  $c_i = \arg \max_k y_{ik}$  is the dominant cell type and  $b_i$  is the entropy quantile bin ( $B$  bins, default  $B = 3$ ).
3. Allocate quota per stratum proportional to its size; sample uniformly within each stratum.

### Strategy 2: Greedy $k$ -Center

1. Pick a random seed point in label space  $\mathbb{R}^K$ .
2. Greedily select the point farthest (in  $\ell_2$ ) from all current centers.
3. Repeat until  $\lfloor \alpha N \rfloor$  points are selected.

**Comparison.** Stratified sampling preserves the joint distribution of cell-type identity and label uncertainty.  $k$ -Center maximizes coverage in label space, ensuring rare cell-type compositions are represented.